`true` if the most recent call to `pred()` returned `true, false` if the time period specified by `relative_time` has elapsed and `pred()` returned `false`.

Note

The potential for spurious wakeups means that it's unspecified how many times `pred` will be called. `pred` will always be invoked with the mutex referenced by `lock` locked, and the function shall return if (and only if) an evaluation of `(bool)pred()` returns `true` or the time period specified by `relative_time` has elapsed. The thread may be blocked for longer than the specified duration. Where possible, the elapsed time is determined by a steady clock.

Throws

Any exception thrown by a call to `pred`, or `std::system_error` if the effects couldn't be achieved.

Synchronization

Calls to `notify_one(), notify_all(), wait(), wait_for()`, and `wait_until()` on a single `std::condition_variable` instance are serialized. A call to `notify_one()` or `notify_all()` will only wake threads that started waiting *prior* to that call.

Std::Condition_Variable::Wait_Until Member Function

Waits until the `std::condition_variable` is notified by a call to `notify_one()` or `notify_all()` or until a specified time has been reached or the thread is woken spuriously.